

Communication connectivity in different solutions-ULTRA



Introduction of Ultra RS485 Port

Ultra's RS485 port is located in COM1 as shown in the following figure.

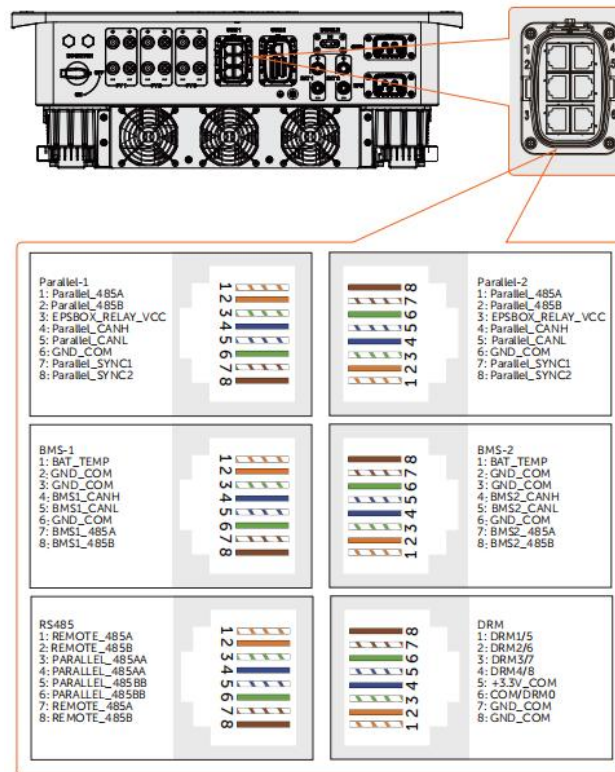


Figure 1 COM 1 Terminal Definition

485 ports are divided into two groups of independent 485 buses. Pins 1, 2, 7 and 8 of port RS485 are used as communication bus and pins 3, 4, 5 and 6 are used as backup bus. For SolaX products, such as the Adapter Box, EV-Charger, and Datahub they can be connected to pin 3 and pin 6 or pin 4 and pin 5. As for pin 1, pin 2, pin 7, and pin 8 they can be utilized to connect devices other than SolaX products. If you require simultaneous connections of multiple devices, a splitter adapter can be employed.

Note:

- The cable length of RS485 communication should not exceed 100 meters.
- Not all devices are compatible with 8 pin network cables. In cases where 8 pin network cables are not supported, it is required to re-crimp the RJ45 terminal according to the pin assignment

Adapter Box G2 Communication connection with Inverter

The inverter communicates with Adapter Box via RS485. In case of excess power, the Adapter Box can utilize it to heat the pump through the connection of dry contacts, SG Ready, or Analog output between the Adapter Box and the heat pump. To power the Adapter Box, an external power adapter is required as the inverter

itself cannot supply power to the Adapter Box. The wiring diagram is shown as follow.

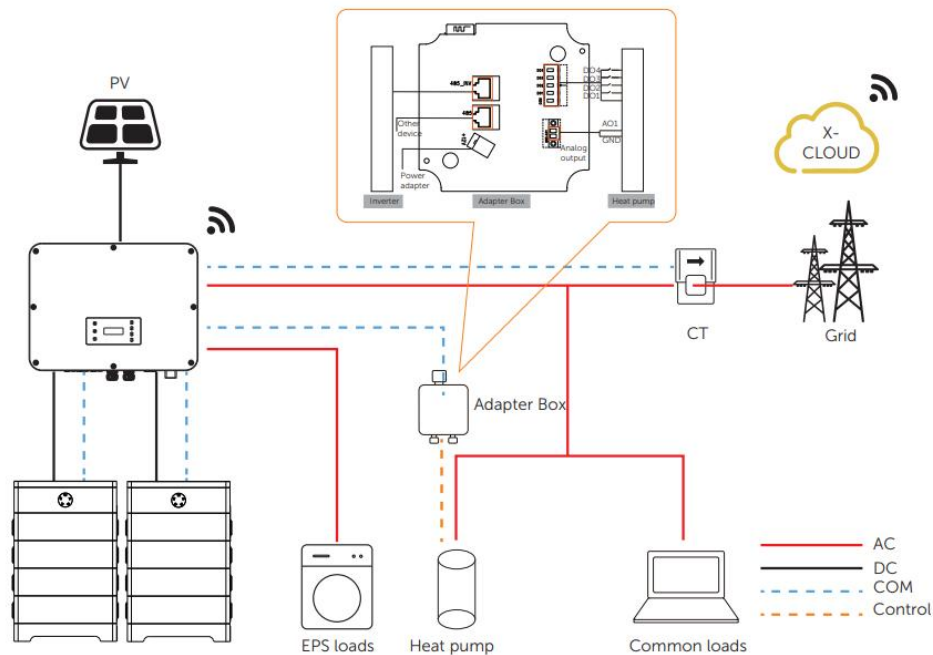


Figure 2 Adapter Box wiring diagram

The communication connection for Adapter Box is shown as follow

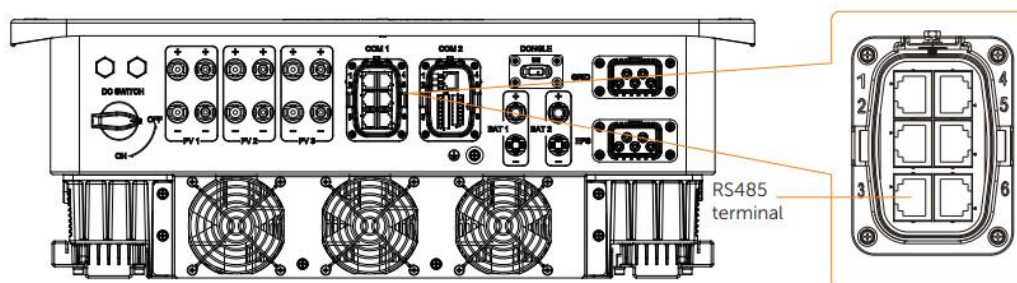


Figure 3 Connection terminal for Adapter Box

Pin-to-pin connection for inverter and Adapter Box G2 is shown as following table

Table 1 Pin-to-pin connection for inverter and Adapter Box G2			
RS485 terminal of inverter		RS485_INV terminal of Adapter Box G2	
Pin	Pin assignment	Pin	Pin assignment
3/4	Parallel_485AA	4	RS485-A
5/6	Parallel_485BB	5	RS485-B

Setting path for Adapter Box G2

Setting path: Menu>Setting>Advance Setting>Modbus

- Select Menu>Setting>Advance Setting>Modbus;
- Select the Adapter Box G2 and set the Baud Rate and corresponding Address.

The default Baud Rate is 9600 and the address is 96;

Note:When two equipments need to be connected at the same time, the band rate and address of the two equipments shall be set to the same.

EV Charger Communication connection with Inverter

The EV-Charger is intended for charging electric vehicles. It should be installed in a fixed location and connected to the AC supply. The EV-Charger can communicate with other devices or systems (inverter, meter, CT, third-party charger management platform, etc.) to realize intelligent control of charging process. The wiring diagram is shown as follow.

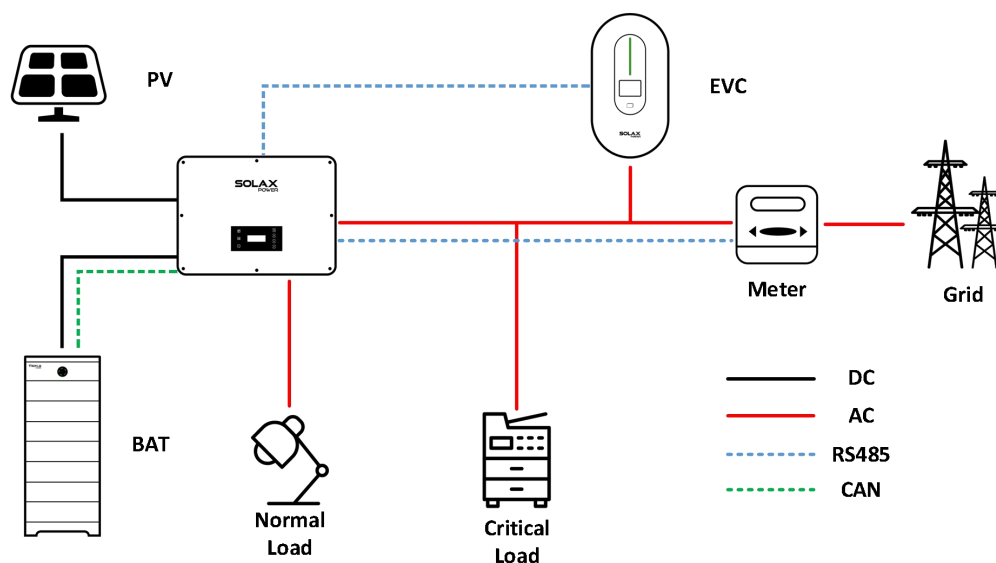


Figure 4 EV Charger wiring diagram

The communication connection for EV Charger is shown as follow

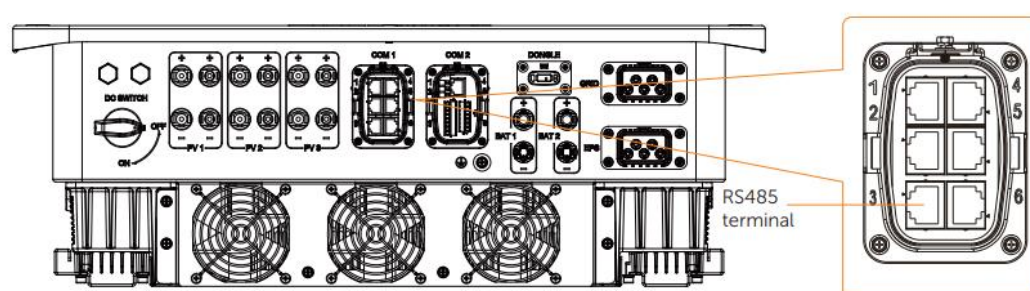


Figure 5 Connection terminal for EV Charger

Pin-to-pin connection for inverter and EV Charger is shown as following table

Table 2 Pin-to-pin connection for inverter and EV Charger			
RS485 terminal of inverter		COM terminal of EV Charger	
Pin	Pin assignment	Pin	Pin assignment
3/4	Parallel_485AA	4	A1
5/6	Parallel_485BB	5	B1

Setting path for EV Charger

Setting path: Menu>Setting>Advance Setting>Modbus

- Select Menu>Setting>Advance Setting>Modbus;
- Select the Adapter Box G2 and set the Baud Rate and corresponding Address.
The default Baud Rate is 9600 and the address is 70;

Note:When two equipments need to be connected at the same time, the baud rate and address of the two equipments shall be set to the same.

Note:When connecting EVC and heat pump, do not use 8pin network cable to plug in, which will affect other pin and cause abnormal communication.

DataHub Communication connection with Inverter

SolaX DataHub can be connected to inverters through RS485 to control the output power of the entire power station according to on-site requirements. Besides, it can work with SolaXCloud to monitor all inverters, allowing for real-time data display and device management. In the entire system, a maximum of 10 X3-ULTRA series inverters can be connected to the DataHub. The wiring diagram as shown as follow.

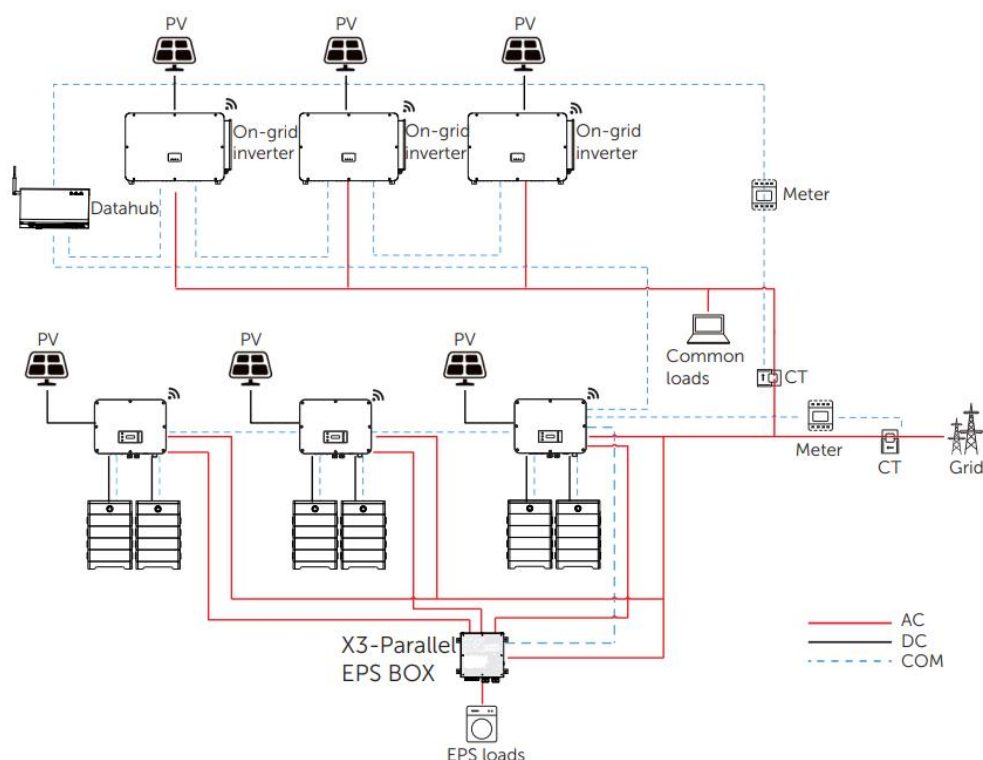


Figure 6 DataHub wiring diagram

The communication connection for Datahub is shown as follow

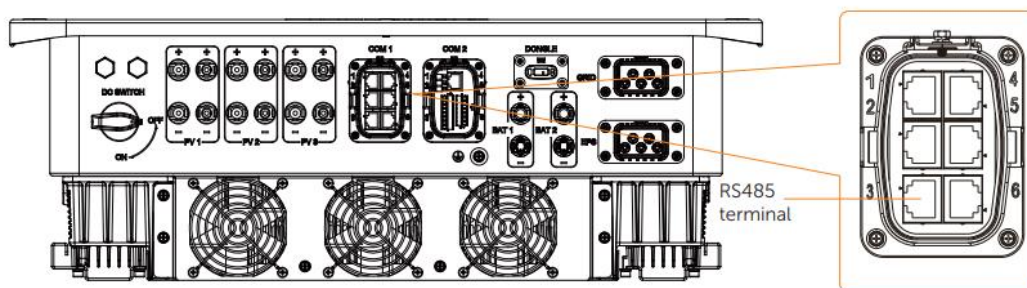
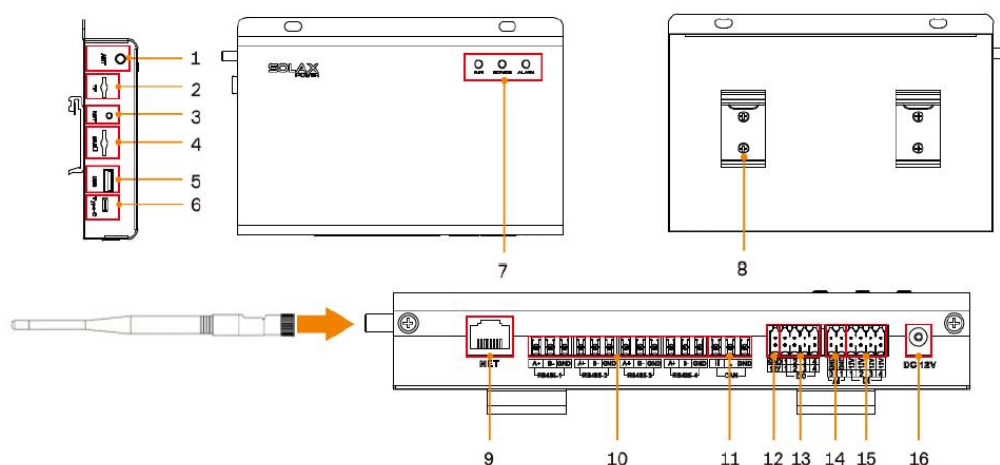


Figure 7 Connection terminal for DataHub

Pin-to-pin connection for inverter and DataHub is shown as follow

Table 3 Pin-to-pin connection for inverter and DataHub

RS485 terminal of inverter		RS485-1 terminal of DataHub	
Pin	Pin assignment	Pin	Pin assignment
3/4	Parallel_485AA	/	A+
5/6	Parallel_485BB	/	B-



- (1) Antenna jack
- (3) RST button (RST)
- (5) USB socket (USB)
- (7) LED indicator (RUN, SERVER, ALARM)
- (9) NET socket (NET)
- (11) CAN socket (CAN)
- (13) DO socket (DO)
- (15) DI socket (DI)

- (2) TF card socket (TF)
- (4) SIM card socket (SIM)
- (6) TYPE-C socket
- (8) Rail clip
- (10) RS485 socket (RS485)
- (12) 12V power output (12V/GND)
- (14) AI socket (AI)
- (16) 12V power input (DC12V)

Setting path for DataHub

Setting path: Menu>Setting>Advance Setting>DataHub

a. Select Menu>Setting>Advance Setting>Modbus;

b. Select the DataHub and set the Baud Rate and corresponding Address.

Note:The braud rate, communication protocol and verification method of the inverters connected to the same RS485 terminal of Datahub must be consistent, and the communication addresses of the inverters must be consecutive and not repeated.